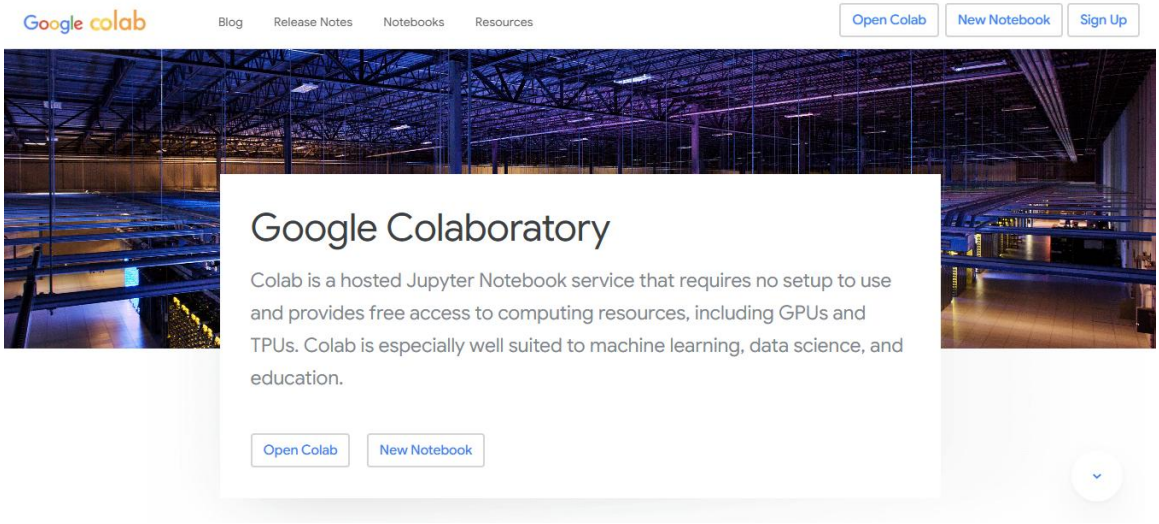
# Guided Tutorial 01: Introduction to Google Colab, Python, AI Libraries, and AI Coding Assistants

This tutorial is designed for beginners who want to:

- Learn how to use Google Colab as a coding environment.
- Get familiar with Python programming.
- Explore popular AI/ML libraries.
- Leverage Google Gemini and GitHub Copilot as AI coding assistants.

By the end, you'll be able to write and run Python code in Colab, understand its environment, and use AI tools to boost your productivity.

## Getting Started with Google Colab




Google Colab (Colaboratory) is a cloud-based Jupyter Notebook environment provided by Google.

Google Colab:

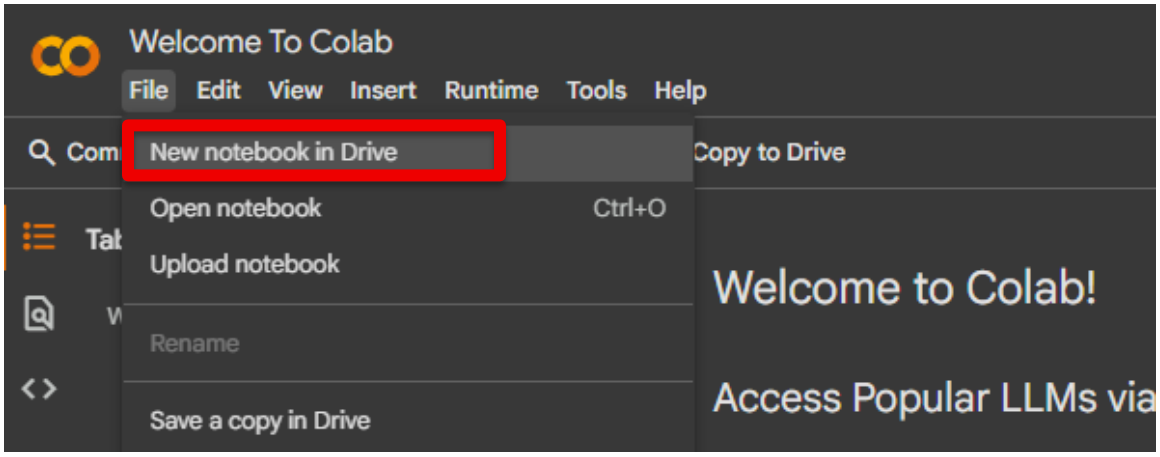
- Runs on the cloud → no installation needed.

- Free access to GPUs and TPUs for AI/ML tasks.
- Allows code + text + visualization in a single document.
- Easily shareable, just like Google Docs.

**Activity:**

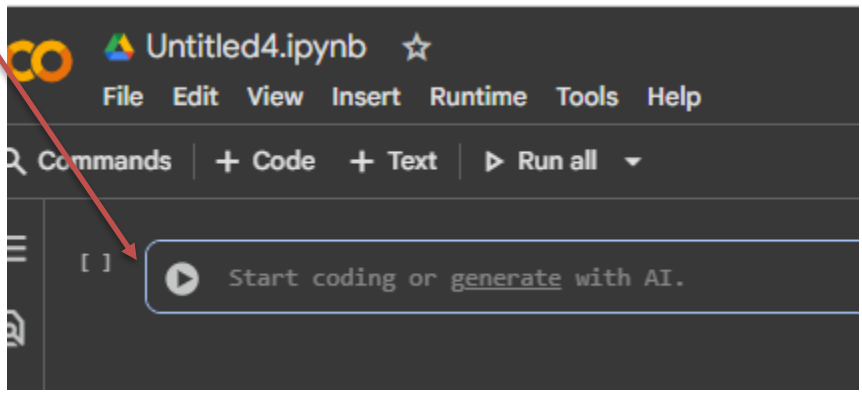
How to Access:

1. Go to https://colab.research.google.com
2. Sign in with your Google Account.
3. Click File → New Notebook to start coding.

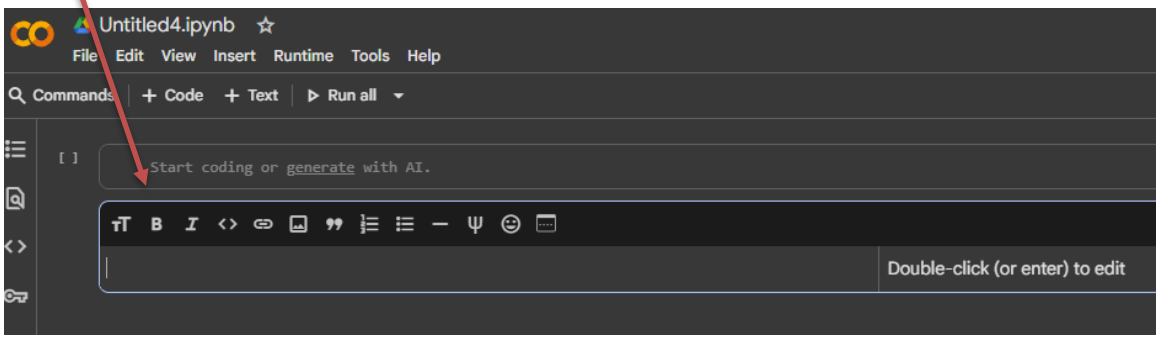



## Exploring the Colab Environment

- Code Cells → where you write Python code.



- Text Cells (Markdown) → add documentation, formulas, or explanations.




- Toolbar → options to run code, add cells, connect to runtime, etc.

- Runtime Types:
  - CPU (default)
  - GPU (for AI/ML tasks)
  - TPU (specialized hardware)

**Activity:**

Change runtime: Runtime → Change runtime type → Select GPU/TPU.

## Python Basics in Colab

**Activity:**

```
print("Hello, Google Colab!")

# Variables
x = 10
y = 3.5
name = "Colab User"
print(x, y, name)
```

```
# Lists & Loops
numbers = [1, 2, 3, 4, 5]
for num in numbers:
    print(num * 2)

Functions:
def greet(name):
    return f"Hello, {name}!"
print(greet("Colab Learner"))
```

## ML Libraries in Python

**NumPy**

**Activity:**

```
import numpy as np
arr = np.array([1, 2, 3])
print(arr * 2)
```

**Pandas**

**Activity:**

```
import pandas as pd
data = {"Name": ["A", "B", "C"], "Score": [85, 90, 78]}
df = pd.DataFrame(data)
print(df)
```

**Matplotlib:**

**Activity:**

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3], [2, 4, 6])
plt.title("Simple Plot")
plt.show()
```

**Scikit-learn:**

**Activity:**

```
from sklearn.linear_model import LinearRegression
import numpy as np

X = np.array([[1], [2], [3]])
y = np.array([2, 4, 6])
model = LinearRegression().fit(X, y)
print("Prediction for 4:", model.predict([[4]]))
```

**TensorFlow:**

**Activity:**

```
import tensorflow as tf
print("TensorFlow version:", tf.__version__)
```

## Using AI Coding Assistants

Following activity use Google Gemini:

**Activity:**

```
# Install client
!pip install google-generativeai

import google.generativeai as genai
genai.configure(api_key="YOUR_API_KEY")

model = genai.GenerativeModel()
response = model.generate_content("Write a Python function to calculate
factorial")
print(response.text)
```

# Hint: How to get Google API Key

